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第一篇 外文资料

Core Banking Systems and Options for Modernization

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原文

Core Banking Systems and Options for Modernization By Julian Alcazar, Sam Baird, Emma Cronenweth, Fumiko Hayashi, and Ken Isaacson Each U.S. depository institution (DI)-including banks and credit unions-uses a back-end information technology system to process daily transactions and manage financial accounts. Many of these “core banking systems” are outdated and unable to fully accommodate modern services, such as open banking and instant payments. Modernizing these systems is a complex process, and DIs may consider a full replacement, a component-based replacement, or augmenting their existing system.

The core banking system of a depository institution (DI)-the back-end information technology system that processes daily banking transactions and updates financial accounts and records -is crucial to the products and services a DI offers its customers. Many DIs still use legacy core systems up to 40 years old that reside on mainframe hardware coded with outdated programming languages (McKinsey & Company 2022). To offer customers better payment services that take full advantage of modern payment methods-such as instant payments-these DIs will need to modernize their core systems.

Despite the importance of modernizing core systems, the systems themselves and the core banking services market are not well understood. Most DIs rely on core banking services providers for their core systems and ancillary services, so modernizing the systems requires close cooperation and coordination with these providers. In addition, ancillary financial services such as payments processing are often integrated with the legacy core systems, making the system architecture complex and difficult to change. This Payments System Research Briefing describes core systems and the options DIs have for modernizing them. Two forthcoming Briefings will examine the market structure surrounding core providers

and explore the role of core providers in facilitating the adoption and broad use of instant payments.

Types of core banking services: Primary or ancillary, in-house or outsourced

Core banking systems provide DIs services that can be categorized broadly as primary or ancillary. Primary services are essential services that the core systems always provide, including account management, customer management, deposit and withdrawal processing, loan processing, and finance and accounting. Ancillary services are additional services the core provider may offer, including payment-related services, bank product interfaces, and customer support. Some DIs outsource all ancillary services to their core provider, while other DIs outsource ancillary services to other providers or perform some of them in-house. Table 1 describes these services in more detail.

Table 1: Core banking services

Service Description Primary Account management Create and manage different types of accounts, such as checking, savings, loans, and credit card accounts

Customer management Create and manage customer profiles, track customer activities, and manage customer relationships

Deposit (credit) and withdrawal Post, process, and manage deposits (credits) to, and withdrawals (debit) processing (debits) from customer accounts

Loan processing Issue, process, and manage various types of loans

Finance and accounting Maintain general ledger, accounts payable, asset and liability management, and credit loss modeling

Ancillary

Payments Authorize and verify funds for outgoing payments (automated clearinghouse, wire, instant payments, cards), capture checks remotely, provide gateway services (or connections) to payment networks/operators, provide ATM management and processing, provide merchant services, and provide enterprise/business account/payment services Bank product interfaces Provide online/mobile banking apps and branch/teller system

Customer support Provide technical support for call center and mobile/online chat

Fraud and risk management Monitor transactions for suspicious activities and issue alerts for various types of bank fraud vulnerabilities Compliance and audit Manage compliance and risk-related functions such as anti-money laundering, countering financing of terrorism

checks, and know-yourcustomer verification Reporting and analytics Generate various reports (for example, balance sheets, income statements, and customer transaction reports) and conduct data analyses to provide insights into customer behavior and market trends

Supporting third-party apps Provide third-party access to customer data and integrate third-party apps (such as Venmo) to online/mobile banking apps

Based on their business needs, DIs employ different combinations of services offered by core providers and other service providers. Larger DIs with ample resources are more likely to use in-house core system processing and solutions, customizing features and functions to meet their specific needs. In contrast, most smaller DIs outsource to core service providers or other service providers, especially in three areas:

IT, business processes, and product offerings (Capgemini 2013). DIs may outsource their IT by using a

third party for services such as server management, infrastructure architecture, and network administration. They may outsource business processes by using a third party to manage an entire business process, such as accounting, finance, or customer support. And they may outsource product

offerings by assigning product development and operations to a third-party provider. Although outsourcing limits DIs' control over some features and functions, it enables them to take advantage of the core (or other service) providers' broad solutions without the need for resource-intensive in-house development or operations.

Legacy core systems versus next generation

Core systems can be categorized as legacy or next generation (ICBA 2021; McKinsey & Company 2022).

Legacy systems typically use a monolithic architecture where all components of the system are

intertwined, potentially in a confusing way due to years of patches and customization. These systems often run on a single server or mainframe, with outdated technology stacks that require specialized skills to maintain. In contrast, next-generation systems use an open, modular architecture, where components are decoupled and can run independently on different servers or in the cloud. These systems use modern technologies such as microservices, application programming interfaces (APIs), and cloud computing.

Next-generation systems have many advantages over legacy systems (McKinsey & Company 2022). Their cloud-native infrastructure typically has lower operating and maintenance costs due to lower reliance on the hardware that may live on a DI's premises. Next-generation systems' plug-and-play features offer flexibility and scalability, increasing access to partnerships with various service providers and enabling a real-time customer experience. Furthermore, next-generation systems enable DIs to easily integrate datasets across the DIs' products and services, giving the DIs new tools and insights into customer behavior and market trends.

The flexibility and scalability of next-generation systems allow DIs to choose whether they pursue a "best-of-breed" strategy or use a fully integrated core system. The best-of-breed strategy enables a DI to pick and choose from vendor solutions that best meet their needs, which may shorten the time required to introduce a new product, such as an instant payment service. Although packaging primary and ancillary services in a next-generation system may not allow a DI to implement a customized feature easily, some DIs may prefer this system due to low internal and external maintenance costs, easy vendor management, and potentially more streamlined operations. Legacy systems may not allow a DI to pursue a best-of-breed strategy due to the technical limitations of integrating newer products into older code.

Despite the numerous advantages of next-generation systems, many DIs have concerns about these systems because they are usually offered by newer, less-proven core providers, such as Finastra, Q2, and Synapse. Lack of experience with next-generation systems and new core providers may increase many DIs' uncertainty about the systems' reliability, security, and ability to meet DIs' needs (ICBA 2021). Incumbent core providers-such as FIS, Fiserv, and Jack Henry and Associates-have long offered legacy core systems but more recently began offering next-generation systems as well. Although these core providers are known for their long track record of dependability, they have been criticized for their lack of services and responsiveness to smaller DIs (ABA 2023; Groenfeldt 2019). The lack of

services and response from their core provider may inhibit an individual DI's ability to meet customers' needs and stay competitive.

Options for DIs to modernize their core systems

Meeting customer demands for modern services-such as open banking, instant payments, and more functional mobile banking-is a significant challenge for DIs with legacy core

systems. However, modernizing these systems is a major endeavor involving extensive planning, expertise, coordination across technology and business domains within their organization, and cooperation with their core provider. The process requires significant time and financial investment, which varies based on approach. Figure 1 depicts the three options DIs have to modernize their core systems: (1) full replacement, (2) component-based replacement, and (3) wrapping or augmenting the existing system.

Figure 1: Three options for DIs to modernize their core system

Sources: Kumar, Salvaterra, and Kulkarni (2019) and authors' modifications.

Option 1: Full replacement

A full replacement is the most risky and expensive option for modernizing a core system and is often the method of last resort when the legacy core system stops meeting business needs (Kumar, Salvaterra, and Kulkarni 2019). The replacement process can introduce many sources of risk, including the downtime of core systems, data migration, the reliability of the new system, and sufficiency of resources to support the transition. Full core conversions can take several years and cost millions (if not hundreds of millions) of dollars depending on the size and complexity of the DI, the scope of implementation, and the deployment approach (BAI 2016). Some industry experts liken this process to “open heart surgery” or “swapping jet engines while flying” (Deloitte 2017; Groenfeldt 2019).

Despite the risk and cost, some DIs have recently chosen to fully replace their core banking systems. Seattle Bank, a single branch bank with \$650 million in assets, decided to fully replace its Fiserv core banking system with a modern, cloud-based core platform from Finastra (Groenfeldt 2019). The decision was driven by the bank's dissatisfaction with the inflexibility and fee structure of their previous legacy

core provider. The new platform includes modern offerings such as enhanced bill pay, banking-as-a-service (BaaS), and open banking tools. 1 Notwithstanding this example, full replacements are rare, particularly among smaller DIs, as they require full commitment from the institution's leadership and careful coordination with the technology team.

Option 2: Component-based replacement

Component-based replacement is less risky than full replacement, in part because DIs replace each component of a core system one at a time (BAI 2016). Disentangling functions of a core system may be difficult for many DIs due to the length of time the system has been in place and the number of layers added over time to keep the system functional. However,

if a DI can separate components within their core system, component-based replacement may be a reasonable modernization strategy.

Some DIs have pursued this option. Zions, a regional bank operating in 11 states with \$90 billion in assets, chose to upgrade the lending component of its core system before moving to the deposit component (Pape 2022). Zions' executive vice president detailed this strategy at the 2022 Digital Banking Conference, noting that the decision to start with the loan system was due to its lower customer engagement and visibility, minimizing potential negative outcomes from conversion issues. After extensively testing with thousands of simulations, Zions then upgraded the deposit component of its core system. Zions ran the old and new systems simultaneously and migrated branches one at a time to mitigate risk. The conversion helped simplify deposit products and streamline processes.

Option 3: Wrapping or augmenting

Instead of replacing a legacy core system, an existing legacy system can sometimes be “wrapped” or augmented with a next-generation parallel “shell” core. Also known as “building on top or the side,” this option involves a next-generation core that plugs into the legacy core via API connections (Sarkar 2021). The next-generation parallel core allows DIs to expand their product offerings, connect to vendors, and remain flexible, while maintaining the integrity of the data and processes of the legacy core system. The next-generation parallel core can process transactions straight through to the legacy core. Because the legacy core and underlying data are maintained, this option is often less risky than replacing components. Although a parallel core solution may be less expensive than a full replacement, DIs may still face costs for integrating services and maintaining multiple core systems over time.

Several new core providers offer next-generation core platforms that wrap or build on top of an existing core, including Finastra, FintechOS, Finzly, Mambu, and SoFi (which acquired Technisys). Using these new core providers may enable DIs to implement instant payments relatively quickly, as some of these companies are the first to provide an API connection to the Federal Reserve's new instant payment service, FedNow (Fitzgerald 2023). These companies also provide an API connection to other payment infrastructures such as Real-Time Payments (RTP), automated clearinghouse (ACH), the Fedwire Funds

Service, the Clearing House Interbank Payments System (CHIPS), and the Society for Worldwide

Interbank Financial Telecommunications (SWIFT). Augmenting a core system with a payment platform that consolidates the processing of multiple payment methods helps DIs improve processing efficiency,

reduce operating costs, and accelerate the time to introduce new services (Baumann and Fishman 2023).

An additional consideration for modernization: Transition to the cloud

No matter which of the three core modernization options DIs choose, they will also face a broader modernization challenge: transitioning to the cloud. Transitioning to the cloud implies that a DI moves existing processes from hardware or mainframes running in DI-managed physical locations to servers hosted by a core provider, vendor, or other third party. This transition can yield simplified hardware maintenance, faster and more flexible data access, faster updates, better scalability, and the ability to integrate with other microservices via APIs (Duepke and others 2019). Moreover, this transition can be pursued concurrently with all three core modernization approaches. For example, DIs may choose a full core system replacement with a cloud-based core provider or upgrade one component at a time to cloud services. DIs may also augment or wrap their legacy core system with services offered by a cloudbased next-generation core provider.

Although many DIs have already started transitioning to the cloud to some degree, they are slow to do so with their core systems. According to a report by Accenture, more than 90 percent of retail banks surveyed around the globe have used the cloud in some capacity, but their adoption for integral services, including core banking services, remains low (Abbott 2021). Influencing this low adoption rate are concerns about data security (and the associated regulatory risks) and outages of cloud service providers. 2 Insufficient transparency from cloud service providers in contract terms (such as data ownership, data back-ups, and data security or privacy provisions) also makes it difficult for DIs to perform necessary due diligence, monitoring, and third-party risk management (U.S. Department of the Treasury 2023). 3

Despite these concerns, some banks have transitioned to the cloud for their core systems. For example, Seattle Bank transitioned to Finastra's cloud-based banking platform in 2020, motivated in part by a plan to integrate with fintechs to provide BaaS solutions and embedded finance (Finastra 2019). 4 One of the benefits gained from transitioning to the cloud was the ability to deploy the Paycheck Protection Program soon after the CARES Act was signed during the pandemic (Seattle Bank 2020). In addition, Mascoma Bank, a DI with

\$2.6 billion of assets, transitioned to Thought Machine for its cloud-native core system (Fintech Futures 2022). This transition integrated previously siloed data and enabled Mascoma to easily implement new product offerings at a faster rate (Thought Machine 2022).

Summary

Core banking systems are the backbone of the products and services that DIs offer their customers. Core systems provide DIs primary services such as account and customer management, deposit and withdrawal processing, loan processing, and finance and accounting. Core systems also often provide ancillary services, such as payment processing and customer support.

While many DIs still use legacy core systems, some have begun modernizing their core systems to meet their customers' demand for modern services, including instant payments. Next-generation systems,

which use modern architecture and technologies, have many advantages over legacy systems and help DIs offer their customers a better banking experience. However, modernizing legacy core systems can be complicated and requires coordination with both core service providers and any other service providers a DI might use.

To modernize legacy core systems, DIs have three options: a full replacement, a component-based replacement, or augmenting the legacy system. The time and financial investment required for modernization and the associated risks vary by option, but all three are a major undertaking. Ultimately, DIs that have already completed their core system modernization and realized the benefits have a competitive advantage in the banking and payments markets.

Endnotes

1 BaaS describes a model in which DIs integrate their digital banking services seamlessly into the products of other

nonbank businesses. Under this model, a DI allows a nonbank business to market the DI's products-such as mobile banking accounts, debit cards, loans, and payment services-under the nonbank business's name. 2 In recent years, two incidents affected confidence in this way. In June 2021, an outage of a distributed denial-of-service

(DDoS) protection service of Akamai Technologies, a U.S.-based cloud company, caused service disruptions at three of the four major Australian banks (Chanthadavong 2021). And

in November and December 2023, about 60 U.S. credit unions experienced outages due to a ransomware attack against Trellance-owned Ongoing Operations, which provides business continuity services and cloud solutions (Birch 2023; Kapko 2023). 3 Ownership of DIs' customer data is also often ambiguous in contract terms between a DI and its core provider. If the core

provider owns the data, the DI will incur sizable cost when switching to another core provider (ICBA 2021). 4 Embedded finance refers to financial solutions that are offered by consumer-facing businesses in conjunction with the

purchase of goods and services. The purpose is to eliminate the need for consumers to leave the merchant's channel to make payments, borrow money, or procure insurance associated with the purchase. BaaS enables embedded finance as well as the provision of financial services by nonbanks (Lehman 2022).

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译文

核心银行系统及其现代化选择

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每一家美国存款机构（DI），包括银行和信用合作社，都使用后端信息技术系统来处理日常交易并管理金融账户。许多此类“核心银行系统”已经过时，无法充分适应开放银行和即时支付等现代服务。对这些系统进行现代化改造是一个复杂过程，存款机构可以考虑完整替换、基于组件的替换，或者在现有系统之上进行封装和增强。

存款机构的核心银行系统，是指用于处理日常银行交易并更新金融账户和记录的后端信息技术系统。它对于存款机构向客户提供的产品和服务至关重要。许多存款机构仍在使用已有几十年历史的遗留核心系统，这些系统往往运行在大型机硬件上，并采用过时的编程语言编写。为了向客户提供更好的支付服务，并充分利用即时支付等现代支付方式，这些机构需要推进核心系统现代化。

尽管核心系统现代化十分重要，但核心系统本身以及核心银行服务市场并不总是被外界充分理解。大多数存款机构依赖核心银行服务提供商来获得核心系统和辅助服务，因此现代化改造需要与这些提供商进行密切合作和协调。此外，支付处理等辅助

金融服务往往已经与遗留核心系统深度集成，使系统架构变得复杂并且难以变更。本文介绍核心系统，并说明存款机构在推进核心系统现代化时可选择的主要路径。

核心银行服务的类型：主要服务与辅助服务、内部自营与外包

核心银行系统向存款机构提供的服务大体可以分为主要服务和辅助服务。主要服务是核心系统必须始终提供的基础服务，包括账户管理、客户管理、存款和取款处理、贷款处理以及财务会计。辅助服务则是核心提供商可能额外提供的服务，包括支付相关服务、银行产品接口和客户支持等。有些存款机构将所有辅助服务都外包给核心提供商；也有些机构将辅助服务外包给其他提供商，或者在机构内部自行完成部分辅助服务。

表 1 核心银行服务

主要服务包括：账户管理，即创建和管理支票、储蓄、贷款、信用卡等不同类型的账户；客户管理，即创建和维护客户档案、跟踪客户活动并管理客户关系；存款（贷记）和取款（借记）处理，即登记、处理并管理客户账户中的存入和支出；贷款处理，即发放、处理和管理不同类型的贷款；财务与会计，即维护总账、应付账款、资产负债管理和信用损失模型。

辅助服务包括：支付服务，即对自动清算所、汇款、即时支付和银行卡等外发支付进行授权和资金验证，远程采集支票，提供连接支付网络或运营方的网关服务，提供自动柜员机管理和处理、商户服务以及企业账户和支付服务；银行产品接口，即提供网上银行、手机银行应用和网点柜员系统；客户支持，即为呼叫中心和移动端或在线聊天提供技术支持；欺诈与风险管理，即监测可疑交易并针对不同类型的银行欺诈漏洞发出警报；合规与审计，即管理反洗钱、反恐融资检查和了解你的客户核验等合规与风险相关功能；报告和分析，即生成资产负债表、利润表、客户交易报告等各种

报告，并开展数据分析以洞察客户行为和市场趋势；支持第三方应用，即向第三方提供客户数据访问能力，并把第三方应用接入网上银行或手机银行应用。

根据自身业务需要，存款机构会采用核心提供商和其他服务提供商提供的不同服务组合。资源充足的大型存款机构更可能采用内部核心系统处理和解决方案，并根据自身特定需求定制功能。相比之下，大多数小型存款机构会外包给核心服务提供商或其他服务提供商，尤其是在信息技术、业务流程和产品供给三个方面。存款机构可以把服务器管理、基础设施架构和网络管理等信息技术服务交给第三方；也可以把会计、财务或客户支持等完整业务流程交给第三方管理；还可以把产品开发和运营分配给第三方提供商。外包虽然会限制存款机构对某些功能的控制，但也使其能够利用核心提供商或其他服务提供商的广泛解决方案，而无需投入大量资源进行内部开发或运营。

遗留核心系统与下一代核心系统

核心系统可以分为遗留系统和下一代系统。遗留系统通常采用单体架构，系统中的所有组件相互缠绕。由于经历了多年补丁修复和定制开发，这种相互依赖关系可能变得混乱。遗留系统往往运行在单一服务器或大型机上，技术栈过时，需要具备专门技能的人员进行维护。相比之下，下一代系统采用开放、模块化架构，各组件彼此解耦，可以在不同服务器或云环境中独立运行。这些系统使用微服务、应用程序编程接口（API）和云计算等现代技术。

下一代系统相对于遗留系统具有许多优势。由于减少了对存款机构本地硬件的依赖，其云原生基础设施通常具有较低的运行和维护成本。下一代系统的即插即用特征带来了灵活性和可扩展性，扩大了与各类服务提供商合作的可能，并能支持实时客户

体验。此外，下一代系统还使存款机构更容易整合跨产品和服务的数据集，从而获得了解客户行为和市场趋势的新工具与新洞察。

下一代系统的灵活性和可扩展性，使存款机构可以选择采用“最佳组合”策略，或者使用完全整合的核心系统。“最佳组合”策略允许存款机构从不同供应商解决方案中选择最适合自身需要的部分，这可能缩短推出即时支付等新产品所需的时间。虽然把主要服务和辅助服务打包到同一个下一代系统中，可能使存款机构不容易实现个性化功能，但部分机构仍会偏好这种系统，因为它的内部和外部维护成本较低，供应商管理更简单，运营也可能更加顺畅。由于把新产品整合到旧代码中存在技术限制，遗留系统可能无法支持存款机构采用“最佳组合”策略。

尽管下一代系统具有很多优势，许多存款机构仍对其有所顾虑，因为这些系统往往由较新的、尚未充分验证的核心提供商提供。缺乏使用下一代系统和新核心提供商的经验，可能增加机构对系统可靠性、安全性以及能否满足自身需求的不确定感。FIS、Fiserv、Jack Henry and Associates 等既有核心提供商长期提供遗留核心系统，近来也开始提供下一代系统。尽管这些提供商以长期可靠的业绩记录而闻名，但也曾因对小型存款机构服务不足、响应不够及时而受到批评。核心提供商的服务和响应不足，可能限制单个存款机构满足客户需求并保持竞争力的能力。

存款机构实现核心系统现代化的选择

对于仍使用遗留核心系统的存款机构来说，满足客户对开放银行、即时支付和功能更完整的手机银行等现代服务的需求，是一项重大挑战。然而，系统现代化是一项大型工程，需要在机构内部的技术领域和业务领域之间进行广泛规划、专业投入和协调，并且需要核心提供商的配合。该过程需要大量时间和资金投入，具体取决于采用

的方法。存款机构实现核心系统现代化主要有三种选择：第一，完整替换；第二，基于组件的替换；第三，封装或增强现有系统。

选择一：完整替换

完整替换是核心系统现代化中风险最高、成本最高的选择，通常是在遗留核心系统已经无法满足业务需求时才采用的最后手段。替换过程会引入许多风险来源，包括核心系统停机、数据迁移、新系统可靠性以及支持过渡所需资源是否充足。完整核心系统转换可能需要数年时间，并可能花费数百万甚至数亿美元，具体取决于存款机构的规模和复杂度、实施范围以及部署方式。一些业内专家把这一过程比作“开胸手术”或“飞行中更换喷气发动机”。

尽管风险和成本都很高，仍有一些存款机构近年选择完全替换核心银行系统。例如，Seattle Bank 是一家单一网点银行，资产规模为 6.5 亿美元，它决定用 Finastra 的现代云端核心平台完整替换原有 Fiserv 核心银行系统。这一决定的部分原因，是该行不满意此前遗留核心提供商的灵活性不足和收费结构。新平台包含增强账单支付、银行即服务（BaaS）以及开放银行工具等现代功能。即便有这样的案例，完整替换仍然很少见，尤其是在小型存款机构中，因为它需要机构领导层的充分承诺，也需要技术团队的精心协调。

选择二：基于组件的替换

基于组件的替换风险低于完整替换，其中一个原因是存款机构可以一次替换核心系统中的一个组件。对于许多存款机构来说，把核心系统的功能拆解开来并不容易，因为该系统已经运行了很长时间，并在多年来叠加了许多用于维持功能的层级。然而，如果机构能够分离核心系统中的组件，基于组件的替换就可能成为一种合理的现代化策略。

一些存款机构已经采用过这种方案。Zions 是一家在美国 11 个州运营、资产规模达 900 亿美元的区域性银行。它选择先升级核心系统中的贷款组件，再转向存款组件。该行执行副总裁在 2022 年数字银行会议上说明了这一策略，指出先从贷款系统开始，是因为该系统与客户直接交互程度和可见性较低，可以降低转换问题产生负面结果的可能。经过数千次模拟的充分测试后，Zions 随后升级了核心系统中的存款组件。为了降低风险，Zions 让旧系统和新系统并行运行，并按网点逐一迁移。此次转换有助于简化存款产品并优化流程。

选择三：封装或增强

除了替换遗留核心系统之外，有时也可以用下一代并行“外壳”核心对现有遗留系统进行“封装”或增强。该方法也被称为“在上方构建”或“在侧面构建”，其核心是让下一代核心通过 API 连接接入遗留核心。下一代并行核心能够使存款机构扩展产品供给、连接供应商并保持灵活性，同时维护遗留核心系统的数据和流程完整性。下一代并行核心可以将交易直通处理到遗留核心。由于遗留核心和底层数据得以保留，这一选择通常比替换组件的风险更低。虽然并行核心解决方案可能低于完整替换的成本，但存款机构仍可能面临服务集成以及长期维护多个核心系统的费用。

多家新的核心提供商提供下一代核心平台，可以在现有核心之上进行封装或构建，包括 Finastra、FintechOS、Finzly、Mambu 以及收购了 Technisys 的 SoFi。使用这些新核心提供商，可能使存款机构较快实现即时支付，因为其中一些公司率先提供了连接美联储新即时支付服务 FedNow 的 API。这些公司还提供连接其他支付基础设施的 API，例如实时支付（RTP）、自动清算所（ACH）、Fedwire 资金服务、清算所银行间支付系统（CHIPS）以及环球银行金融电信协会（SWIFT）。用能够整合多

种支付方式处理的平台增强核心系统，有助于存款机构提高处理效率、降低运营成本，并加快推出新服务的速度。

现代化的另一项考虑：向云迁移

无论存款机构选择上述三种核心现代化方案中的哪一种，它们都还会面临更广泛的现代化挑战：向云迁移。向云迁移意味着存款机构把原本运行在本机构管理的物理位置中的硬件或大型机上的既有流程，迁移到由核心提供商、供应商或其他第三方托管的服务器上。该迁移可以带来更简化的硬件维护、更快速和灵活的数据访问、更快的更新、更好的可扩展性，以及通过 API 与其他微服务集成的能力。此外，向云迁移可以与三种核心现代化路径同时推进。例如，存款机构可以选择由云端核心提供商完成完整核心系统替换，也可以一次将一个组件升级到云服务。存款机构还可以利用基于云的下一代核心提供商提供的服务，对遗留核心系统进行增强或封装。

虽然许多存款机构已经在某种程度上开始迁移到云，但它们在核心系统方面的迁移速度较慢。Accenture 的一份报告显示，全球受访零售银行中有 90% 以上已经以某种方式使用云，但包括核心银行服务在内的关键服务采用云的程度仍然较低。影响这一低采用率的因素包括对数据安全及相关监管风险的担忧，以及对云服务提供商中断的担忧。云服务提供商合同条款中的透明度不足，例如数据所有权、数据备份、数据安全或隐私条款，也会使存款机构难以开展必要的尽职调查、监控和第三方风险管理。

尽管存在这些担忧，一些银行已经把核心系统迁移到云。例如，Seattle Bank 于 2020 年迁移到 Finastra 的云端银行平台，其中一个动机是计划与金融科技企业集成，以提供银行即服务解决方案和嵌入式金融。迁移到云之后，该行获得的一项收益是，在疫情期间 CARES 法案签署后不久，就能够部署薪资保护计划。此外，资产规模为

26 亿美元的 Mascoma Bank 迁移到了 Thought Machine 的云原生核心系统。这一迁移整合了此前彼此孤立的数据，使 Mascoma 能够更快地轻松推出新产品。

总结

核心银行系统是存款机构向客户提供产品和服务的骨干。核心系统向存款机构提供账户和客户管理、存款和取款处理、贷款处理以及财务会计等主要服务。核心系统也经常提供支付处理和客户支持等辅助服务。

尽管许多存款机构仍在使用遗留核心系统，一些机构已经开始对核心系统进行现代化改造，以满足客户对即时支付等现代服务的需求。采用现代架构和技术的下一代系统相对于遗留系统具有许多优势，并有助于存款机构向客户提供更好的银行体验。然而，遗留核心系统现代化可能非常复杂，需要与核心服务提供商以及存款机构可能使用的其他服务提供商进行协调。

为了实现遗留核心系统现代化，存款机构有三种选择：完整替换、基于组件的替换，或增强遗留系统。不同方案所需的时间和资金投入以及伴随的风险各不相同，但三种方案都属于重大工程。最终，那些已经完成核心系统现代化并实现收益的存款机构，将在银行和支付市场中拥有竞争优势。

注释

1. BaaS 描述的是一种模式，即存款机构把自身数字银行服务无缝集成到非银行企业的产品之中。在这种模式下，存款机构允许非银行企业以该企业自身名义营销存款机构的产品，例如手机银行账户、借记卡、贷款和支付服务。

2. 近年发生的两起事件影响了机构对云服务的信心。2021 年 6 月，美国云公司 Akamai Technologies 的分布式拒绝服务（DDoS）防护服务发生中断，导致澳大利亚四大主要银行中的三家出现服务故障。2023 年 11 月和 12 月，美国约 60 家信用合作

社由于针对 Trellance 旗下 Ongoing Operations 的勒索软件攻击而发生服务中断，该公司提供业务连续性服务和云解决方案。

3. 在存款机构与核心提供商之间的合同条款中，存款机构客户数据的所有权也常常不够明确。如果核心提供商拥有数据，存款机构在切换到另一家核心提供商时将产生相当高的成本。

4. 嵌入式金融是指面向消费者的企业在销售商品和服务的同时提供金融解决方案。其目的在于，使消费者无需离开商户渠道即可完成付款、借款或购买与交易相关的保险。BaaS 不仅支持嵌入式金融，也支持非银行机构提供金融服务。

参考文献

原文参考文献保留英文格式，见前文原文部分。

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本科毕业设计 (论文)

外文资料翻译

毕业设计题目	智能客服机器人的设计 与交互优化
外文资料题目	Artificial Intelligence Risk Management Framework (AI RMF 1.0) (节选)
学 院	计算机科学与技术学院
专 业	软件工程
学 生 姓 名	王奇
指导教师姓名	谢海明，魏昱恒

第二篇 外文资料

Artificial Intelligence Risk Management Framework (AI RMF

1.0) (节选)

资料来源：Elham Tabassi. Artificial Intelligence Risk Management Framework (AI RMF 1.0). NIST AI 100-1, National Institute of Standards and Technology, January 2023. 本文节选 Executive Summary 与 3. AI Risks and Trustworthiness，译文仅用于课程作业，非官方译本。

原文

Executive Summary

Artificial intelligence (AI) technologies have significant potential to transform society and people's lives - from commerce and health to transportation and cybersecurity to the environment and our planet. AI technologies can drive inclusive economic growth and support scientific advancements that improve the conditions of our world. AI technologies, however, also pose risks that can negatively impact individuals, groups, organizations, communities, society, the environment, and the planet. Like risks for other types of technology, AI risks can emerge in a variety of ways and can be characterized as long- or short-term, high or low-probability, systemic or localized, and high- or low-impact. The AI RMF refers to an AI system as an engineered or machine-based system that can, for a given set of objectives, generate outputs such as predictions, recommendations, or decisions influencing real or virtual environments. AI systems are designed to operate with varying levels of autonomy (Adapted from: OECD Recommendation on AI:2019; ISO/IEC 22989:2022). While there are myriad standards and best practices to help organizations mitigate the risks of traditional software or information-based systems, the risks posed by AI systems are in many ways unique (See Appendix B). AI systems, for example, may be trained on data that can change over time, sometimes significantly and unexpectedly, affecting system functionality and trustworthiness in ways that are hard to understand. AI systems and the contexts in which they are deployed are frequently complex, making it difficult to detect and respond to failures when they occur. AI systems are inherently socio-technical in nature, meaning they are influenced by societal dynamics and human behavior. AI risks - and benefits - can emerge from the interplay of technical aspects combined with

societal factors related to how a system is used, its interactions with other AI systems, who operates it, and the social context in which it is deployed. These risks make AI a uniquely challenging technology to deploy and utilize both for organizations and within society. Without proper controls, AI systems can amplify, perpetuate, or exacerbate inequitable or undesirable outcomes for individuals and communities. With proper controls, AI systems can mitigate and manage inequitable outcomes. AI risk management is a key component of responsible development and use of AI systems. Responsible AI practices can help align the decisions about AI system design, development, and uses with intended aim and values. Core concepts in responsible AI emphasize human centricity, social responsibility, and sustainability. AI risk management can drive responsible uses and practices by prompting organizations and their internal teams who design, develop, and deploy AI to think more critically about context and potential or unexpected negative and positive impacts. Understanding and managing the risks of AI systems will help to enhance trustworthiness, and in turn, cultivate public trust.

Social responsibility can refer to the organization's responsibility "for the impacts of its decisions and activities on society and the environment through transparent and ethical behavior" (ISO 26000:2010). Sustainability refers to the "state of the global system, including environmental, social, and economic aspects, in which the needs of the present are met without compromising the ability of future generations to meet their own needs" (ISO/IEC TR 24368:2022). Responsible AI is meant to result in technology that is also equitable and accountable. The expectation is that organizational practices are carried out in accord with "professional responsibility," defined by ISO as an approach that "aims to ensure that professionals who design, develop, or deploy AI systems and applications or AI-based products or systems, recognize their unique position to exert influence on people, society, and the future of AI" (ISO/IEC TR 24368:2022). As directed by the National Artificial Intelligence Initiative Act of 2020 (P.L. 116-283), the goal of the AI RMF is to offer a resource to the organizations designing, developing, deploying, or using AI systems to help manage the many risks of AI and promote trustworthy and responsible development and use of AI systems. The Framework is intended to be voluntary, rights-preserving, non-sector-specific, and use-case agnostic, providing flexibility to organizations of all sizes and in all sectors and throughout society to implement the approaches in the Framework. The Framework is designed to equip organizations and individuals - referred to here as AI actors - with approaches that increase the trustworthiness of AI systems, and to help foster the responsible design, development, deployment, and use of AI systems over time. AI actors

are defined by the Organisation for Economic Co-operation and Development (OECD) as “those who play an active role in the AI system lifecycle, including organizations and individuals that deploy or operate AI” [OECD (2019) Artificial Intelligence in Society-OECD iLibrary] (See Appendix A). The AI RMF is intended to be practical, to adapt to the AI landscape as AI technologies continue to develop, and to be operationalized by organizations in varying degrees and capacities so society can benefit from AI while also being protected from its potential harms. The Framework and supporting resources will be updated, expanded, and improved based on evolving technology, the standards landscape around the world, and AI community experience and feedback. NIST will continue to align the AI RMF and related guidance with applicable international standards, guidelines, and practices. As the AI RMF is put into use, additional lessons will be learned to inform future updates and additional resources. The Framework is divided into two parts. Part 1 discusses how organizations can frame the risks related to AI and describes the intended audience. Next, AI risks and trustworthiness are analyzed, outlining the characteristics of trustworthy AI systems, which include

valid and reliable, safe, secure and resilient, accountable and transparent, explainable and interpretable, privacy enhanced, and fair with their harmful biases managed. Part 2 comprises the “Core” of the Framework. It describes four specific functions to help organizations address the risks of AI systems in practice. These functions - GOVERN,

MAP, MEASURE, and MANAGE - are broken down further into categories and subcategories. While GOVERN applies to all stages of organizations’ AI risk management processes and procedures, the MAP, MEASURE, and MANAGE functions can be applied in AI

system-specific contexts and at specific stages of the AI lifecycle. Additional resources related to the Framework are included in the AI RMF Playbook, which is available via the NIST AI RMF website: <https://www.nist.gov/itl/ai-risk-management-framework>. Development of the AI RMF by NIST in collaboration with the private and public sectors is directed and consistent with its broader AI efforts called for by the National AI Initiative Act of 2020, the National Security Commission on Artificial Intelligence recommendations, and the Plan for Federal Engagement in Developing Technical Standards and Related Tools. Engagement with the AI community during this Framework’s development

- via responses to a formal Request for Information, three widely attended workshops, public comments on a concept paper and two drafts of the Framework, discussions at multiple public forums, and many small group meetings - has informed development of the

AI RMF 1.0 as well as AI research and development and evaluation conducted by NIST and others. Priority research and additional guidance that will enhance this Framework will be captured in an associated AI Risk Management Framework Roadmap to which NIST and the broader community can contribute.

3. AI Risks and Trustworthiness

For AI systems to be trustworthy, they often need to be responsive to a multiplicity of criteria that are of value to interested parties. Approaches which enhance AI trustworthiness can reduce negative AI risks. This Framework articulates the following characteristics of trustworthy AI and offers guidance for addressing them. Characteristics of trustworthy AI systems include: valid and reliable, safe, secure and resilient, accountable and transparent, explainable and interpretable, privacy-enhanced, and fair with harmful bias managed. Creating trustworthy AI requires balancing each of these characteristics based on the AI system's context of use. While all characteristics are socio-technical system attributes, accountability and transparency also relate to the processes and activities internal to an AI system and its external setting. Neglecting these characteristics can increase the probability and magnitude of negative consequences.

Fig. 4. Characteristics of trustworthy AI systems. Valid & Reliable is a necessary condition of

trustworthiness and is shown as the base for other trustworthiness characteristics. Accountable & Transparent is shown as a vertical box because it relates to all other characteristics. Trustworthiness characteristics (shown in Figure 4) are inextricably tied to social and organizational behavior, the datasets used by AI systems, selection of AI models and algorithms and the decisions made by those who build them, and the interactions with the humans who provide insight from and oversight of such systems. Human judgment should be employed when deciding on the specific metrics related to AI trustworthiness characteristics and the precise threshold values for those metrics. Addressing AI trustworthiness characteristics individually will not ensure AI system trustworthiness; tradeoffs are usually involved, rarely do all characteristics apply in every setting, and some will be more or less important in any given situation. Ultimately, trustworthiness is a social concept that ranges across a spectrum and is only as strong as its weakest characteristics. When managing AI risks, organizations can face difficult decisions in balancing these characteristics. For example, in certain scenarios tradeoffs may emerge between optimizing for interpretability and achieving privacy. In other cases, organizations might face a tradeoff

between predictive accuracy and interpretability. Or, under certain conditions such as data sparsity, privacy-enhancing techniques can result in a loss in accuracy, affecting decisions

about fairness and other values in certain domains. Dealing with tradeoffs requires taking into account the decision-making context. These analyses can highlight the existence and extent of tradeoffs between different measures, but they do not answer questions about how to navigate the tradeoff. Those depend on the values at play in the relevant context and should be resolved in a manner that is both transparent and appropriately justifiable. There are multiple approaches for enhancing contextual awareness in the AI lifecycle. For example, subject matter experts can assist in the evaluation of TEVV findings and work with product and deployment teams to align TEVV parameters to requirements and deployment conditions. When properly resourced, increasing the breadth and diversity of input from interested parties and relevant AI actors throughout the AI lifecycle can enhance opportunities for informing contextually sensitive evaluations, and for identifying AI system benefits and positive impacts. These practices can increase the likelihood that risks arising in social contexts are managed appropriately. Understanding and treatment of trustworthiness characteristics depends on an AI actor's particular role within the AI lifecycle. For any given AI system, an AI designer or developer may have a different perception of the characteristics than the deployer. Trustworthiness characteristics explained in this document influence each other. Highly secure but unfair systems, accurate but opaque and uninterpretable systems, and inaccurate but secure, privacy-enhanced, and transparent systems are all undesirable. A comprehensive approach to risk management calls for balancing tradeoffs among the trustworthiness characteristics. It is the joint responsibility of all AI actors to determine whether AI technology is an appropriate or necessary tool for a given context or purpose, and how to use it responsibly. The decision to commission or deploy an AI system should be based on a contextual assessment of trustworthiness characteristics and the relative risks, impacts, costs, and benefits, and informed by a broad set of interested parties.

3.1 Valid and Reliable

Validation is the “confirmation, through the provision of objective evidence, that the requirements for a specific intended use or application have been fulfilled” (Source: ISO 9000:2015). Deployment of AI systems which are inaccurate, unreliable, or poorly generalized to data and settings beyond their training creates and increases negative AI risks and reduces trustworthiness. Reliability is defined in the same standard as the “ability of an

item to perform as required, without failure, for a given time interval, under given conditions” (Source: ISO/IEC TS 5723:2022). Reliability is a goal for overall correctness of AI system operation under the conditions of expected use and over a given period of time, including the entire lifetime of the system.

Accuracy and robustness contribute to the validity and trustworthiness of AI systems, and can be in tension with one another in AI systems. Accuracy is defined by ISO/IEC TS 5723:2022 as “closeness of results of observations, computations, or estimates to the true values or the values accepted as being true.” Measures of accuracy should consider computational-centric measures (e.g., false positive and false negative rates), human-AI teaming, and demonstrate external validity (generalizable beyond the training conditions). Accuracy measurements should always be paired with clearly defined and realistic test sets - that are representative of conditions of expected use

- and details about test methodology; these should be included in associated documentation. Accuracy measurements may include disaggregation of results for different data

segments. Robustness or generalizability is defined as the “ability of a system to maintain its level of performance under a variety of circumstances” (Source: ISO/IEC TS 5723:2022). Robustness is a goal for appropriate system functionality in a broad set of conditions and circumstances, including uses of AI systems not initially anticipated. Robustness requires not only that the system perform exactly as it does under expected uses, but also that it should perform in ways that minimize potential harms to people if it is operating in an unexpected setting. Validity and reliability for deployed AI systems are often assessed by ongoing testing or monitoring that confirms a system is performing as intended. Measurement of validity, accuracy, robustness, and reliability contribute to trustworthiness and should take into consideration that certain types of failures can cause greater harm. AI risk management efforts should prioritize the minimization of potential negative impacts, and may need to include human intervention in cases where the AI system cannot detect or correct errors.

3.2 Safe

AI systems should “not under defined conditions, lead to a state in which human life, health, property, or the environment is endangered” (Source: ISO/IEC TS 5723:2022). Safe operation of AI systems is improved through:

- responsible design, development, and deployment practices;

- clear information to deployers on responsible use of the system;
- responsible decision-making by deployers and end users; and
- explanations and documentation of risks based on empirical evidence of incidents.

Different types of safety risks may require tailored AI risk management approaches based on context and the severity of potential risks presented. Safety risks that pose a potential risk of serious injury or death call for the most urgent prioritization and most thorough risk management process.

Employing safety considerations during the lifecycle and starting as early as possible with planning and design can prevent failures or conditions that can render a system dangerous. Other practical approaches for AI safety often relate to rigorous simulation and in-domain testing, real-time monitoring, and the ability to shut down, modify, or have human intervention into systems that deviate from intended or expected functionality. AI safety risk management approaches should take cues from efforts and guidelines for safety in fields such as transportation and healthcare, and align with existing sector- or application-specific guidelines or standards.

3.3 Secure and Resilient

AI systems, as well as the ecosystems in which they are deployed, may be said to be resilient if they can withstand unexpected adverse events or unexpected changes in their environment or use - or if they can maintain their functions and structure in the face of internal and external change and degrade safely and gracefully when this is necessary (Adapted from: ISO/IEC TS 5723:2022). Common security concerns relate to adversarial examples, data poisoning, and the exfiltration of models, training data, or other intellectual property through AI system endpoints. AI systems that can maintain confidentiality, integrity, and availability through protection mechanisms that prevent unauthorized access and use may be said to be secure. Guidelines in the NIST Cybersecurity Framework and Risk Management Framework are among those which are applicable here. Security and resilience are related but distinct characteristics. While resilience is the ability to return to normal function after an unexpected adverse event, security includes resilience but also encompasses protocols to avoid, protect against, respond to, or recover from attacks. Resilience relates to robustness and goes beyond the provenance of the data to encompass unexpected or adversarial use (or abuse or misuse) of the model or data.

3.4 Accountable and Transparent

Trustworthy AI depends upon accountability. Accountability presupposes transparency. Transparency reflects the extent to which information about an AI system and its outputs is available to individuals interacting with such a system - regardless of whether they are even aware that they are doing so. Meaningful transparency provides access to appropriate levels of information based on the stage of the AI lifecycle and tailored to the role or knowledge of AI actors or individuals interacting with or using the AI system. By promoting higher levels of understanding, transparency increases confidence in the AI system. This characteristic's scope spans from design decisions and training data to model training, the structure of the model, its intended use cases, and how and when deployment, post-deployment, or end user decisions were made and by whom. Transparency is often necessary for actionable redress related to AI system outputs that are incorrect or otherwise lead to negative impacts. Transparency should consider human-AI interaction: for exam-

ple, how a human operator or user is notified when a potential or actual adverse outcome caused by an AI system is detected. A transparent system is not necessarily an accurate, privacy-enhanced, secure, or fair system. However, it is difficult to determine whether an opaque system possesses such characteristics, and to do so over time as complex systems evolve. The role of AI actors should be considered when seeking accountability for the outcomes of AI systems. The relationship between risk and accountability associated with AI and technological systems more broadly differs across cultural, legal, sectoral, and societal contexts. When consequences are severe, such as when life and liberty are at stake, AI developers and deployers should consider proportionally and proactively adjusting their transparency and accountability practices. Maintaining organizational practices and governing structures for harm reduction, like risk management, can help lead to more accountable systems. Measures to enhance transparency and accountability should also consider the impact of these efforts on the implementing entity, including the level of necessary resources and the need to safeguard proprietary information. Maintaining the provenance of training data and supporting attribution of the AI system's decisions to subsets of training data can assist with both transparency and accountability. Training data may also be subject to copyright and should follow applicable intellectual property rights laws. As transparency tools for AI systems and related documentation continue to evolve, developers of AI systems are encouraged to test different types of transparency tools in cooperation with AI deployers to ensure that AI systems are used as intended.

3.5 Explainable and Interpretable

Explainability refers to a representation of the mechanisms underlying AI systems' operation, whereas interpretability refers to the meaning of AI systems' output in the context of their designed functional purposes. Together, explainability and interpretability assist those operating or overseeing an AI system, as well as users of an AI system, to gain deeper insights into the functionality and trustworthiness of the system, including its outputs. The underlying assumption is that perceptions of negative risk stem from a lack of ability to make sense of, or contextualize, system output appropriately. Explainable and interpretable AI systems offer information that will help end users understand the purposes and potential impact of an AI system. Risk from lack of explainability may be managed by describing how AI systems function, with descriptions tailored to individual differences such as the user's role, knowledge, and skill level. Explainable systems can be debugged and monitored more easily, and they lend themselves to more thorough documentation, audit, and governance.

Risks to interpretability often can be addressed by communicating a description of why an AI system made a particular prediction or recommendation. (See "Four Principles of Explainable Artificial Intelligence" and "Psychological Foundations of Explainability and Interpretability in Artificial Intelligence" found [here](#).)

Transparency, explainability, and interpretability are distinct characteristics that support each other. Transparency can answer the question of "what happened" in the system. Explainability can answer the question of "how" a decision was made in the system. Interpretability can answer the question of "why" a decision was made by the system and its meaning or context to the user.

3.6 Privacy-Enhanced

Privacy refers generally to the norms and practices that help to safeguard human autonomy, identity, and dignity. These norms and practices typically address freedom from intrusion, limiting observation, or individuals' agency to consent to disclosure or control of facets of their identities (e.g., body, data, reputation). (See The NIST Privacy Framework: A Tool for Improving Privacy through Enterprise Risk Management.) Privacy values such as anonymity, confidentiality, and control generally should guide choices for AI system design, development, and deployment. Privacy-related risks may influence security, bias, and transparency and come with tradeoffs with these other characteristics. Like safety and security, specific technical features of an AI system may promote or reduce privacy. AI

systems can also present new risks to privacy by allowing inference to identify individuals or previously private information about individuals. Privacy-enhancing technologies (“PETs”) for AI, as well as data minimizing methods such as de-identification and aggregation for certain model outputs, can support design for privacy-enhanced AI systems. Under certain conditions such as data sparsity, privacyenhancing techniques can result in a loss in accuracy, affecting decisions about fairness and other values in certain domains.

3.7 Fair - with Harmful Bias Managed

Fairness in AI includes concerns for equality and equity by addressing issues such as harmful bias and discrimination. Standards of fairness can be complex and difficult to define because perceptions of fairness differ among cultures and may shift depending on application. Organizations’ risk management efforts will be enhanced by recognizing and considering these differences. Systems in which harmful biases are mitigated are not necessarily fair. For example, systems in which predictions are somewhat balanced across demographic groups may still be inaccessible to individuals with disabilities or affected by the digital divide or may exacerbate existing disparities or systemic biases.

Bias is broader than demographic balance and data representativeness. NIST has identified three major categories of AI bias to be considered and managed: systemic, computational and statistical, and human-cognitive. Each of these can occur in the absence of prejudice, partiality, or discriminatory intent. Systemic bias can be present in AI datasets, the organizational norms, practices, and processes across the AI lifecycle, and the broader society that uses AI systems. Computational and statistical biases can be present in AI datasets and algorithmic processes, and often stem from systematic errors due to non-representative samples. Human-cognitive biases relate to how an individual or group perceives AI system information to make a decision or fill in missing information, or how humans think about purposes and functions of an AI system. Human-cognitive biases are omnipresent in decision-making processes across the AI lifecycle and system use, including the design, implementation, operation, and maintenance of AI. Bias exists in many forms and can become ingrained in the automated systems that help make decisions about our lives. While bias is not always a negative phenomenon, AI systems can potentially increase the speed and scale of biases and perpetuate and amplify harms to individuals, groups, communities, organizations, and society. Bias is tightly associated with the concepts of transparency as well as fairness in society. (For more information about bias, including the

three categories, see NIST Special Publication 1270, Towards a Standard for Identifying and Managing Bias in Artificial Intelligence.)

译文

人工智能风险管理框架 (AI RMF 1.0) (节选)

资料节选范围：Executive Summary；3. AI Risks and Trustworthiness。

执行摘要

人工智能 (AI) 技术具有显著潜力，可以改变社会和人们的生活。这种影响覆盖商业、健康、交通、网络安全、环境以及整个地球。AI 技术能够推动包容性经济增长，并支持科学进步，从而改善世界的运行条件。然而，AI 技术也会带来风险，这些风险可能对个人、群体、组织、社区、社会、环境和地球产生负面影响。与其他技术风险一样，AI 风险可能以多种方式出现，并可被描述为长期或短期、高概率或低概率、系统性或局部性、高影响或低影响。

在 AI RMF 中，AI 系统是指工程化的或基于机器的系统。为了给定的一组目标，AI 系统能够生成预测、建议或决策等输出，并影响真实或虚拟环境。AI 系统被设计为能够以不同程度的自主性运行。

虽然已经存在大量标准和最佳实践，帮助组织缓解传统软件或信息系统的风险，但 AI 系统带来的风险在许多方面具有独特性。例如，AI 系统可能使用会随时间变化的数据进行训练，而且这种变化有时显著且难以预期，从而以不易理解的方式影响系统功能和可信度。AI 系统及其部署环境往往非常复杂，一旦发生故障，检测和响应都会比较困难。AI 系统本质上具有社会技术属性，也就是说，它们受到社会动态和人类行为的影响。AI 风险和收益可能来自技术因素与社会因素之间的相互作用，这

些社会因素包括系统如何被使用、系统与其他 AI 系统如何交互、由谁来操作系统，以及系统部署所在的社会背景。

这些风险使 AI 成为一种在组织和社会中部署与使用时都极具挑战性的技术。如果缺乏适当控制，AI 系统可能放大、延续或加剧对个人和社区不公平或不理想的结果。通过适当控制，AI 系统则可以缓解和管理不公平结果。

AI 风险管理是负责任地开发和使用 AI 系统的关键组成部分。负责任 AI 实践有助于使关于 AI 系统设计、开发和使用的决策，与预期目标 and 价值保持一致。负责任 AI 的核心概念强调以人为本、社会责任和可持续性。AI 风险管理能够推动负责任的使用和实践，因为它促使设计、开发和部署 AI 的组织及其内部团队，更批判性地思考情境以及潜在或意外的负面和正面影响。理解并管理 AI 系统风险，有助于增强可信度，并进一步培育公众信任。

社会责任可以指组织通过透明和合乎伦理的行为，对其决策和活动对社会与环境造成的影响承担责任。可持续性指全球系统的状态，包括环境、社会和经济方面，在这种状态下，当代人的需求得到满足，同时不损害后代满足其自身需求的能力。负责任 AI 意味着技术也应当是公平且可问责的。组织实践应当符合“职业责任”的要求，即设计、开发或部署 AI 系统、AI 应用或基于 AI 的产品与系统的专业人员，应认识到自身对人、社会和 AI 未来具有影响力的特殊位置。

根据《2020 年国家人工智能倡议法案》的要求，AI RMF 的目标是为设计、开发、部署或使用 AI 系统的组织提供一种资源，帮助它们管理 AI 的多重风险，并促进可信赖、负责任的 AI 系统开发和使用。该框架旨在保持自愿性、权利保护性、非部门特定性和用例无关性，从而为不同规模、不同部门以及整个社会中的组织实施框架方法提供灵活性。

该框架旨在为组织和个人提供方法，以提高 AI 系统的可信度，并推动 AI 系统在整个生命周期中的负责任设计、开发、部署和使用。本文把这些组织和个人称为 AI 参与者。经济合作与发展组织将 AI 参与者定义为“在 AI 系统生命周期中发挥积极作用的主体，包括部署或运行 AI 的组织和个人”。

AI RMF 旨在具有实用性，能够随着 AI 技术持续发展而适应 AI 格局，并能够由不同能力和程度的组织进行操作化，使社会既能从 AI 中受益，又能受到保护，免受其潜在伤害。

该框架及其配套资源将根据技术演进、全球标准格局以及 AI 社群的经验和反馈持续更新、扩展和改进。NIST 将继续使 AI RMF 及相关指南与适用的国际标准、指南和实践保持协调。随着 AI RMF 投入使用，还将积累更多经验，为未来更新和补充资源提供依据。

该框架分为两个部分。第一部分讨论组织如何界定与 AI 相关的风险，并描述预期受众。随后，框架分析 AI 风险与可信度，概述可信 AI 系统的特征，包括有效且可靠、安全、稳健且有弹性、可问责且透明、可说明且可解释、隐私增强，以及通过管理有害偏见来实现公平。

第二部分构成框架的“核心”。它描述四项具体功能，用以帮助组织在实践中应对 AI 系统风险。这些功能为治理 (GOVERN)、映射 (MAP)、测量 (MEASURE) 和管理 (MANAGE)，并进一步划分为类别和子类别。治理功能适用于组织 AI 风险管理流程和程序的所有阶段；映射、测量和管理功能则可应用于特定 AI 系统情境以及 AI 生命周期的特定阶段。

与框架相关的其他资源包含在 AI RMF 行动手册中。行动手册可通过 NIST AI RMF 网站获得。

NIST 与私营和公共部门合作开发 AI RMF，这项工作受《2020 年国家人工智能倡议法案》、国家人工智能安全委员会建议以及联邦参与制定技术标准和相关工具计划等更广泛 AI 工作的指导并与之保持一致。在框架开发过程中，AI 社群通过对正式信息征集的回应、三次广泛参与的研讨会、对概念文件和两个框架草案的公开评论、多次公共论坛讨论以及许多小组会议参与其中。这些互动为 AI RMF 1.0 的形成以及 NIST 和其他机构开展的 AI 研发与评估提供了依据。未来还将通过相关的 AI 风险管理框架路线图，记录能够增强本框架的优先研究和补充指南，NIST 及更广泛的社群都可以为此作出贡献。

3. AI 风险与可信度

要使 AI 系统值得信任，它们通常需要回应多个对相关方有价值的标准。增强 AI 可信度的方法可以降低负面 AI 风险。本框架阐明可信 AI 的若干特征，并提供处理这些特征的指导。可信 AI 系统的特征包括：有效且可靠、安全、稳健且有弹性、可问责且透明、可说明且可解释、隐私增强，以及公平且有害偏见得到管理。构建可信 AI 需要根据 AI 系统的使用情境，在这些特征之间进行平衡。虽然所有特征都是社会技术系统属性，但可问责性和透明性还涉及 AI 系统内部及其外部环境中的过程和活动。忽视这些特征，会增加负面后果发生的概率和严重程度。

图 4 说明了可信 AI 系统的特征。有效且可靠是可信度的必要条件，被视为其他可信度特征的基础。可问责且透明以纵向框的形式呈现，因为它与其他所有特征相关。

可信度特征与社会和组织行为、AI 系统使用的数据集、AI 模型与算法的选择、构建者所作出的决策，以及对系统提供洞察和监督的人类之间的互动密不可分。在决

定与 AI 可信度特征有关的具体指标，以及这些指标的精确阈值时，应当运用人的判断。

单独处理 AI 可信度的每一项特征，并不能确保 AI 系统整体可信。通常会涉及取舍，很少有所有特征都适用于每一种场景，而且在给定情境中，一些特征会比另一些更重要。归根结底，可信度是一种社会概念，它存在于一个连续谱上，并且其强度只取决于最薄弱的特征。

在管理 AI 风险时，组织可能会在平衡这些特征时面临困难决策。例如，在某些场景中，优化可解释性与实现隐私之间可能出现取舍。在其他情况下，组织可能需要在预测准确性和可解释性之间权衡。又如，在数据稀疏等特定条件下，隐私增强技术可能造成准确性损失，进而影响某些领域中关于公平和其他价值的决策。处理这些取舍需要考虑决策情境。相关分析能够揭示不同测量指标之间是否存在取舍以及取舍程度，但并不能直接回答应如何在取舍中作出选择。如何选择取决于相关情境中的价值因素，并应以透明且适当可辩护的方式加以解决。

在 AI 生命周期中有多种方法可以增强情境意识。例如，主题专家可以协助评估测试、评价、验证和确认 (TEVV) 的结果，并与产品和部署团队合作，使 TEVV 参数与需求和部署条件保持一致。在资源充分的情况下，在 AI 生命周期中扩大相关方和 AI 参与者输入的广度与多样性，可以增加开展情境敏感型评估的机会，也有助于识别 AI 系统收益和正面影响。这些实践能够提高组织适当管理社会情境中产生的风险的可能性。

对可信度特征的理解和处理，取决于 AI 参与者在 AI 生命周期中的具体角色。对于同一个 AI 系统，AI 设计者或开发者对这些特征的理解，可能不同于部署者。

本文说明的可信度特征彼此相互影响。高度安全但不公平的系统、准确但不透明且不可解释的系统，以及不准确但安全、隐私增强且透明的系统，都是不可取的。全

面的风险管理方法要求在可信度特征之间平衡取舍。所有 AI 参与者都共同负责判断 AI 技术对于给定情境或目的是否适当或必要，以及应如何负责任地使用它。是否委托开发或部署某个 AI 系统，应建立在对可信度特征以及相关风险、影响、成本和收益的情境化评估之上，并应吸收广泛相关方的信息。

3.1 有效且可靠

验证是指“通过提供客观证据，确认已经满足特定预期用途或应用的要求”。部署不准确、不可靠，或者无法很好泛化到训练数据和训练环境之外的数据与场景的 AI 系统，会产生并增加负面 AI 风险，并降低可信度。

可靠性是指“在给定条件下，在给定时间间隔内，某项事物按要求无故障运行的能力”。可靠性是 AI 系统在预期使用条件下、在给定时间内实现整体正确运行的目标，该时间范围包括系统的整个生命周期。

准确性和鲁棒性有助于 AI 系统的有效性和可信度，但二者在 AI 系统中也可能相互张力。准确性是指观察、计算或估计结果与真实值或被接受为真实的值之间的接近程度。准确性测量应当考虑以计算为中心的指标，例如假阳性率和假阴性率；也应考虑人机协作，并证明外部有效性，即能够泛化到训练条件之外。准确性测量应始终配合明确界定且现实的测试集，这些测试集应代表预期使用条件，并应说明测试方法；这些内容应包含在相关文档中。准确性测量还可以对不同数据片段的结果进行拆分呈现。

鲁棒性或泛化能力是指系统“在多种环境条件下维持其性能水平的能力”。鲁棒性的目标，是使系统在广泛条件和情境下都能具备适当功能，包括 AI 系统最初未预料到的用途。鲁棒性不仅要求系统在预期用途下表现一致，还要求系统在意外环境中运行时，能够以尽量减少对人的潜在伤害的方式运行。

已经部署的 AI 系统的有效性和可靠性，通常通过持续测试或监控来评估，以确认系统是否按预期运行。对有效性、准确性、鲁棒性和可靠性的测量有助于可信度建设，并应考虑某些类型的故障可能造成更大伤害。AI 风险管理工作应优先减少潜在负面影响；在 AI 系统无法检测或纠正错误时，可能需要纳入人工干预。

3.2 安全

AI 系统在规定条件下不应导致人的生命、健康、财产或环境处于危险状态。AI 系统的安全运行可以通过以下方式改进：负责任的设计、开发和部署实践；向部署者提供关于系统负责任使用的清晰信息；部署者和最终用户作出负责任的决策；以及基于事件经验证据对风险进行说明和记录。

不同类型的安全风险，可能需要根据背景和潜在风险严重程度采取定制化的 AI 风险管理方法。对于可能造成严重伤害或死亡的安全风险，应给予最紧急的优先级，并采用最充分的风险管理流程。

在生命周期中纳入安全考虑，并尽可能从规划和设计早期开始，可以防止系统失效或出现使系统危险的条件。AI 安全的其他实践方法通常包括严格仿真和领域内测试、实时监控，以及在系统偏离预期或设计功能时能够关闭、修改系统或引入人工干预。

AI 安全风险管理工作应借鉴交通、医疗等领域关于安全的工作和指南，并与现有行业或应用特定指南、标准保持一致。

3.3 稳健且有弹性

如果 AI 系统以及其部署所在的生态系统能够承受意外不利事件，或环境与使用方式中的意外变化，或者能够在内部和外部变化面前维持其功能与结构，并在必要时以安全、平稳的方式降级，那么可以认为其具有弹性。常见安全关切包括对抗样本、

数据投毒，以及通过 AI 系统端点窃取模型、训练数据或其他知识产权。能够通过防止未经授权访问和使用的保护机制，维持机密性、完整性和可用性的 AI 系统，可以被认为是安全的。NIST 网络安全框架和风险管理框架中的指南均可适用于此。

安全性和弹性相关但并不相同。弹性是指在意外不利事件后恢复正常功能的能力；安全性包含弹性，但还包括用于避免、抵御、响应或从攻击中恢复的协议。弹性与鲁棒性有关，但超越数据来源问题，还涵盖模型或数据的意外使用、对抗性使用、滥用或误用。

3.4 可问责且透明

可信 AI 依赖可问责性，而可问责性以透明性为前提。透明性反映有关 AI 系统及其输出的信息，在多大程度上可供与该系统互动的个人获得，不论这些个人是否意识到自己正在与该系统互动。具有意义的透明性，应根据 AI 生命周期阶段，并结合 AI 参与者或与 AI 系统互动、使用 AI 系统的个人的角色和知识水平，提供适当层级的信息访问。透明性通过促进更高水平的理解，增强人们对 AI 系统的信心。

这一特征的范围从设计决策和训练数据，到模型训练、模型结构、预期用例，以及部署、部署后或最终用户决策如何、何时、由谁作出。对于那些错误或以其他方式造成负面影响的 AI 系统输出，透明性通常是实现可行动救济的必要条件。透明性还应考虑人机交互。例如，当检测到由 AI 系统造成的潜在或实际不利结果时，人类操作者或用户如何收到通知。透明的系统并不必然准确、隐私增强、安全或公平；但对于不透明系统来说，判断其是否具备这些特征，以及随着复杂系统演化持续进行这种判断，都会更加困难。

在追究 AI 系统结果的责任时，应考虑 AI 参与者的角色。AI 及更广泛技术系统所关联的风险与问责之间的关系，会随着文化、法律、行业和社会背景而不同。当后

果严重，例如生命和自由受到影响时，AI 开发者和部署者应当按比例、主动地调整其透明性和问责实践。维持用于减少伤害的组织实践和治理结构，例如风险管理，有助于形成更可问责的系统。

增强透明性和问责性的措施，还应考虑这些努力对实施主体的影响，包括所需资源水平以及保护专有信息的需要。

维护训练数据来源，并支持把 AI 系统决策归因到训练数据的子集，有助于透明性和问责性。训练数据也可能受到版权保护，因此应遵守适用的知识产权法律。

随着 AI 系统透明性工具和相关文档不断发展，鼓励 AI 系统开发者与 AI 部署者合作测试不同类型的透明性工具，以确保 AI 系统按预期方式使用。

3.5 可说明且可解释

可说明性指对 AI 系统运行机制的表征；可解释性则指 AI 系统输出在其设计功能目的背景下的含义。二者共同帮助运行或监督 AI 系统的人，以及 AI 系统用户，更深入地理解系统的功能和可信度，包括理解系统输出。这里的基本假设是，负面风险感知源于人们无法恰当理解系统输出或把系统输出置于情境之中。可说明且可解释的 AI 系统能够提供信息，帮助最终用户理解 AI 系统的目的和潜在影响。

缺乏可说明性带来的风险，可以通过描述 AI 系统如何运行来管理，并且这种描述应根据用户角色、知识和技能水平等个体差异进行调整。可说明的系统更容易调试和监控，也更适合形成充分的文档、审计和治理。

可解释性风险通常可以通过说明 AI 系统为何作出某项预测或建议来处理。

透明性、可说明性和可解释性是相互支持但彼此不同的特征。透明性可以回答系统中“发生了什么”的问题。可说明性可以回答系统中的决策“如何”形成。可解释性可以回答系统“为何”作出某项决策，以及该决策对用户的含义或上下文。

3.6 隐私增强

隐私一般指有助于保护人的自主性、身份和尊严的规范与实践。这些规范和实践通常涉及免受侵扰、限制观察，或者个人同意披露或控制身份不同侧面（如身体、数据和声誉）的能力。

匿名性、保密性和控制等隐私价值，通常应指导 AI 系统设计、开发和部署的选择。与隐私相关的风险可能影响安全、偏见和透明性，也可能与这些特征之间产生取舍。与安全性和网络安全一样，AI 系统的具体技术特征可能促进隐私，也可能削弱隐私。AI 系统还可能带来新的隐私风险，例如通过推断识别个人或获取此前属于个人私密的信息。

AI 的隐私增强技术（PETs），以及针对某些模型输出的数据最小化方法，例如去标识化和聚合，都可以支持隐私增强型 AI 系统设计。在数据稀疏等特定条件下，隐私增强技术可能造成准确性损失，进而影响某些领域中有关公平和其他价值的决策。

3.7 公平：有害偏见得到管理

AI 中的公平包括对平等与公正的关注，重点处理有害偏见和歧视等问题。公平标准可能复杂且难以定义，因为不同文化对公平的理解不同，并且这种理解可能会随应用场景而变化。组织在风险管理工作中识别并考虑这些差异，将有助于增强风险管理效果。有害偏见得到缓解的系统并不必然就是公平的。例如，某些系统的预测结果可能在不同人口群体之间大体平衡，但仍可能让残障人士难以使用，或受到数字鸿沟影响，或者加剧既有差距和系统性偏见。

偏见的范围比人口统计平衡和数据代表性更广。NIST 识别出三类需要考虑和管理的主要 AI 偏见：系统性偏见、计算与统计偏见，以及人类认知偏见。即使不存在

偏见、偏袒或歧视意图，这三类偏见也可能出现。系统性偏见可能存在于 AI 数据集、AI 生命周期中的组织规范、实践和流程，以及使用 AI 系统的更广泛社会之中。计算与统计偏见可能存在于 AI 数据集和算法过程中，常常源于非代表性样本所导致的系统性错误。人类认知偏见则与个人或群体如何理解 AI 系统信息、如何据此作出决策或填补缺失信息，以及人类如何看待 AI 系统的目的和功能有关。人类认知偏见存在于 AI 生命周期和系统使用中的各类决策过程中，包括 AI 的设计、实施、运行和维护。

偏见具有多种形式，并可能嵌入帮助我们对生活作出决策的自动化系统中。偏见并不总是负面现象，但 AI 系统可能提高偏见传播的速度和规模，并延续、放大对个人、群体、社区、组织和社会的伤害。偏见与透明性以及社会公平概念紧密相关。

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本科毕业设计 (论文)

外文资料翻译

毕业设计题目	智能客服机器人的设计 与交互优化
外文资料题目	Artificial Intelligence and Machine Learning in Financial Services (节选)
学 院	计算机科学与技术学院
专 业	软件工程
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第三篇 外文资料

Artificial Intelligence and Machine Learning in Financial Services (节选)

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原文

Artificial Intelligence and Machine Learning in Financial Services

The financial industry's adoption of artificial intelligence (AI) and machine learning (ML) is evolving as financial firms employ ever greater levels of technology and automation to deliver services. Expanding on earlier models of quantitative analysis, AI/ML has often been adopted in finance to solve discrete challenges, such as maximizing profit and minimizing risk. Yet the industry's adoption of the newer technology also occurs against perceptions that are steeped in tradition and historical financial regulation, and regulators want to ensure that the technology does not sidestep regulations frequently described as technology neutral.

Technological advances in computer hardware, capacity, and data storage—which permit the collection and analysis of data—helped fuel the development and use of AI/ML technologies in finance. Unlike older algorithms that automated human-coded rules, new AI models can learn by themselves and make inferences and recommendations not identified by modelers in advance. This shift in technology has also enabled the use of new types of data including alternative data, unstructured data, and unlabeled information data—which, when combined, extend the technologies' uses to new financial services or products.

Different parts of the financial services industry have adopted AI/ML technology to varying degrees and for various purposes. Some uses of AI/ML include powering chatbots in customer service functions, identifying investment opportunities and/or executing trades, augmenting lending models or making lending decisions, and identifying and preventing fraud. The extent to which a sector or firm adopts various technologies reflects a variety of factors, including a firm's ability to fund internal development and regulatory requirements.

Financial industry policymakers face competing pressures. Financial service providers and technology companies are likely to continue adopting and promoting AI/ML to save time and money and promote accessibility, accuracy, and regulatory compliance. However, challenges and risks in the form of bias, potential for systemic risk and manipulation, affordability, and consequences for employment remain. Determining whether the existing regulatory structure is sufficient—or whether one that is more closely tailored to the technological capacities of the evolving technology is necessary—has emerged as a key consideration. Should Congress consider the legislative framework governing AI/ML in finance, industry and consumers alike will expect that it weighs the benefits of innovation with existing and potential future challenges and risks.

Introduction

Artificial intelligence (AI) is the general term used to describe the process of programming computers and machines to think and operate like humans. Machine learning (ML) is a subset of AI that describes computers and programs that may be programmed to operate with minimal human intervention and can in some instances learn and/or update themselves. Various events over the past few years have helped raise the profile of AI/ML and its role in delivering financial services. OpenAI's introduction of the large language model (LLM) ChatGPT in 2022 was a rare moment when an AI/ML technology became directly accessible by the broad public. While momentum around AI/ML had been building over the past decade at least, the ubiquity of competing prominent technologies concentrated attention on all manner of applications, including those in the financial sector.

AI/ML is accelerating a long, ongoing shift in finance from face-to-face interactions conducted in a customer's community to online products and services fueled by advanced algorithms that require little or no human interaction and can occur anywhere. Such applications may have benefits for financial institutions and their clients. Relative to human operation or models built using earlier technology, AI/ML provision of financial services may be faster and cheaper due to their speed and efficiency, and they may be able to expand service to more individuals due to their ability to analyze alternative data and find latent connections. The financial industry's use of technology is not new, and there are elements to its adoption of AI/ML that resemble past periods.

Adoption of the technology also brings challenges and risks. While AI may eliminate some human bias in financial services, it may also introduce or exacerbate bias. Certain AI-market specific factors—such as using the same data to train models and concentration of technology and technological capacity—may encourage monocultures, or herd-like behavior, that introduce systemic risk. The ability of certain models to learn on their own, free from human oversight, may also allow them to engage in market manipulation, conflicts of interest, or other unlawful activity.

This report discusses the role of AI/ML in financial services broadly. First, it provides a brief overview of some technical terms. Next, it discusses the motivations for the financial industry's use of AI/ML. It then provides an overview of some financial applications of the technology. Finally, it addresses the regulatory framework and considers various policy issues, including any applicable regulations.

Technology Overview in Brief

AI is the broad label for technologies that give computer systems the ability to learn new concepts or tasks and to reason and draw useful conclusions about the world. Earlier versions of AI were largely dependent on their human developers that programmed machines with a series of deterministic rules, such as if-then statements, alongside troves of subject matter content. Such technology required specific syntax for questions or commands and produced responses on topics for which it had been programmed. When faced with a situation for which it had not been programmed, it would freeze. That dependency would likely disqualify it from the AI in current taxonomy and is largely disappearing. According to some, most applications of AI now are ML, so the terms as used are synonymous.

ML is a subset of AI focused on constructing computer systems that can automatically improve through experiencing and determining certain characteristics inherent in all learning systems, which can include computers, humans, and organizations. ML attempts to give computers the ability to learn and change without being reprogrammed. It uses algorithms—sequenced actions designed to solve specific problems—that improve automatically with various degrees of, and often little or no, human interaction. In other words, unlike earlier AI, it is meant to be adaptable. ML systems are intended to determine relationships among variables or recognize patterns in large data sets. Yet ML is not one thing, such as a discrete computer application, nor is it even

one technology. Instead, it is a group of systems that is defined in part by the type of information used to train the model and the amount of involvement provided by human trainers.

Technological developments in computing and telecommunications have been a boon to AI/ML, and the financial industry's use of it includes advanced performance computing, which includes hardware, cloud and edge computing, and communication technologies. However, the financial industry's adoption of AI/ML depends upon the availability of data, including alternative data not traditionally used in financial decisionmaking, and the ability to process and analyze unstructured data. These technological advances move in tandem with the amassing of large quantities of data that require scalable architecture for efficient storage, manipulation, and analysis.

Some of today's technologies have their conceptual origins decades ago, and they are now being developed. Unsupervised learning's ability to process unlabeled data may identify relationships—for example, in market interactions or among loan applications—where its programmers had not thought to look. Similarly, the ability to process unstructured and alternative data that previously could not be analyzed—such as images and social media posts—has opened it to analysis.

Big Data

Big data refers both to a type of information and how it is used. It refers to large data sets that exhibit specific characteristics such as volume, velocity, variety, and/or variability and requires scalable architecture to store, manage, and analyze. The term also generally refers to the data use by managers, companies, and industries to facilitate business decisions.

Some advances in ML have been directly attributed to the ability to process more data. The specific big data used in finance depends on the sector or use case. For example, some firms may use big data—including social media—when deciding whether to make loans. In a capital markets and trading environment, big data may include continuous data feeds from various stock exchanges, which can include stock prices and broader market moves and histories of transactions. Data in this environment may also include company-level data that may influence a stock performance, including financial statements and quarterly and annual regulatory filings. Alternative data refers to information that has not traditionally been used to make financial decisions. In consumer credit underwriting, alternative data refers to non-credit data that has not traditionally been used in determining credit scores, such as rent and utility payments. Banks and

credit card and payment companies may also use alternative and unstructured data—such as transaction information and location or GPS data—in their fraud detection and anti-money laundering activities. Asset managers and other capital markets participants may also use GPS and satellite imagery of retailer parking lots or social media mentions that provide insight into a company's performance. The availability of different forms of data and the models they use have expanded the ways financial firms may use both.

Recent interest in AI has been marked by a period of increased investment in and development and adoption of AI/ML technologies by various sectors, including the financial industry.

According to one bank's research, investment in AI is expected to climb to around \$100 billion in the United States in 2025 and nearly \$200 billion globally. Notably, there is likely broad investment from traditional financial institutions, with finance-focused startups attracting investment from venture capital firms.

Finance and Artificial Intelligence

According to the Organisation for Economic Co-operation and Development (OECD), AI is powering digital transformation that has been rapidly deployed in finance. Adoption of this newer technology is not surprising as finance has often been linked with cutting-edge technology. However, the industry is also known for its use of often obsolete technology and associations with traditional practices that are at odds with new technologies. While in the past, personal banking was rooted in community relationships and conducted in person, now nearly all transactions can be performed online. In capital markets transactions, electronic trading has essentially ended the need for trading to be conducted person to person at the physical location of exchanges. Meanwhile, the use of AI/ML, like the related adoption of a broad set of financial technology, also has implications for regulation, which prioritizes industry transparency, explainability, and fairness. Finance's adoption of AI/ML thus takes place as technology and its capacity may be evolving more rapidly than societal and regulatory expectations.

Motivation behind the financial sector's adoption of AI is not new. Finance's adoption of AI has been compared to the development of net-present-value calculations, the invention of the automated teller machine, use of expert systems for personal financial planning, and automated trading, all of which employed different technologies.

In addition to a general gravitation to technology, various characteristics of the financial industry have made it a ripe testing ground for AI/ML. Information processing is the central function of

financial markets. For example, AI/ML generally requires large quantities of data on which to train its systems. Data is a quintessential feature across the various financial sectors. The industry amasses data from economic indicators, financial markets, individual consumers and businesses, and payments and transactions, among countless other sources. Moreover, while comprised of different subsectors—with diverse functions, services, and products—finance presents quantifiable problems, the solutions for which are often reduced to maximizing returns considering a given risk tolerance while considering some number of other quantifiable variables. As such, the industry is generating large amounts of data and providing the operating environment for AI/ML technologies.

AI/ML may also increase the speed with which financial transactions may take place. For example, high-frequency trading, which is also increasingly automated and programmatic, allows trades to occur in the smallest fractions of a second, speeds at which neither human analysis nor execution are possible. Also, if a financial institution uses AI/ML to determine whether it should make a loan, the speed may be evident in multiple ways. AI/ML-induced speed may allow financial institutions to update their lending models. Lenders may also use AI/ML-based lending models, integrated with digital platforms and web interfaces, to deliver decisions to customers more quickly.

AI/ML Applications in Finance

Financial industry and market participants use AI for a variety of functions. These functions have multiple purposes, including improving efficiency, assisting with decisionmaking, producing analysis and forecasting, increasing profitability, managing risk, and underwriting credit. Moreover, they may be used in support functions or front office functions. Back office applications may include post-trade processing, trading profit and loss reconciliations, and data analytics. Other applications may include regulatory compliance functions, such as know-your-customer checks and customer identification programs, anti-money laundering and countering the financing of terrorism programs, and anti-fraud. AI may also assist in asset allocation, robo-advising, and trade execution. Many of the actions taken by financial firms are multistep and closely related, and AI may be used for and benefit from different parts of a business or phases of a business process.

Credit underwriting—assessing the risk of prospective borrowers defaulting on loan repayment—is one of banks' and other financial institutions' primary businesses. Loan-making financial

institutions have automated the methods for making loan decisions, including through the use of electronic data. For consumer loans, such methods generally entail using a mathematical formula called a scoring model to determine a consumer's credit score and subsequently whether the firm should make the loan and at what interest rate. Various factors—including bill paying history, unpaid debt, outstanding loans, and accounts—go into determining a credit score. Recently, however, some financial companies have considered using AI/ML to augment or replace traditional credit scoring.

Lenders have been using predictive models for decades. Traditionally, statistical regression methods used data from the credit reporting bureaus, assigning weights to different variables to help forecast whether an applicant would default on a loan and to determine the likelihood of on-time loan repayment. More recently, firms may use ML-based measures based on their ability to analyze large amounts and different types of data—including transaction data—and their ability to discern other important relationships not visible in traditional models. Adoption of AI/ML for credit underwriting may differ among banks and nonbank fintech. Various regulatory requirements may make banks hesitant to use the technology for credit underwriting decisions. Generally, because nonbanks may have different regulatory requirements and supervision than banks do, they may be more willing to use AI/ML technology.

Chatbots are computer programs that interact with people online by simulating human conversation through text and voice commands. Banking chatbots provide immediate assistance 24/7, reducing wait times, addressing customer inquiries, providing information on account balances and transaction history, and guiding users through various banking processes. Chatbots used in banking provide substantial cost savings annually, according to one estimate.

Regtech, or regulatory technology, refers to regulated financial institutions' use of technology to meet various regulatory, compliance, and data reporting functions. Regtech has existed for around a decade. It had focused on compliance with regulations related to onboarding and identifying new customers, but more recently it has been described as being instrumental in anti-money laundering, countering the financing of terrorism, fraud prevention, risk management, stress testing, and micro and macroprudential reporting. For example, banks and other financial institutions use robotics process automation to help them comply with reporting requirements and repopulating required data on a regular basis. In many cases, banks' and other financial institutions' adoption of regtech may involve partnering with third-party providers with expertise in the field.

A key form of regtech aided by AI/ML is detecting, preventing, and reporting unauthorized and illicit financial activities for banks and other financial institutions. The shift to AI/ML solutions is due to its more adaptable approaches and its ability to leverage more types of data. Banks and other financial institutions must detect, prevent, and report on unauthorized and illicit financial activity, and there has been an increase in the number of financial institutions using AI/ML to address the issue. Banks and other financial institutions can train models on huge volumes of consumer behavior data they generate, allowing the ML models to learn fraud patterns and to then detect fraudulent behavior in practice. Similarly, there may be benefits to using the technology across the anti-money laundering value chain, including at onboarding and client screening and with particular and immediate benefits coming from transaction monitoring. In addition, researchers suggest that the technology is also useful in reducing false positives, freeing banks to dedicate resources to actual instances of fraud.

Natural language processing (NLP) is a branch of AI that trains computers to understand and process language as humans do. NLP can be used to develop LLMs that can create content in the form of essays or computer code from prompts using techniques that predict the next word in a sequence of words based on certain probabilities. This technology supports applications in finance, including powering chatbots, surveying companies' own large troves of data, and writing code. LLMs are examples of artificial neural networks that attempt to reflect the functioning of the human brain by replicating neurons as mathematical representations and organizing layers of such artificial neurons into clusters or layers. Aside from finance-specific applications, LLMs and ChatGPT in particular provided general public audiences with some of the first concrete and directly applicable examples of AI that until now had been reserved for specialists.

中文

金融服务中的人工智能与机器学习（节选）

资料节选范围：Introduction；Technology Overview in Brief；Finance and Artificial Intelligence；AI/ML Applications in Finance。

金融服务中的人工智能与机器学习

金融行业对人工智能（AI）和机器学习（ML）的采用正在不断发展。金融机构在提供服务时使用越来越高水平的技术和自动化工具。AI/ML 延续了早期定量分析模型的思路，常被用于解决相对明确的金融问题，例如最大化收益和最小化风险。不过，金融行业采用新技术的过程，也处在传统观念和历史金融监管框架之中。监管者希望确保这些技术不会绕开通常被描述为技术中立的监管要求。

计算机硬件、计算能力和数据存储方面的技术进步，使数据收集和分析成为可能，也推动了 AI/ML 技术在金融领域的发展。与早期自动执行人工编码规则的算法不同，新的 AI 模型能够自行学习，并作出建模人员事先没有明确设定的推断和建议。这种技术变化也使金融机构能够使用新的数据类型，包括替代数据、非结构化数据和无标签信息数据。它们结合在一起，可以把相关技术扩展到新的金融服务或产品之中。

金融服务行业的不同领域，以不同程度、出于不同目的采用 AI/ML 技术。AI/ML 的应用包括驱动客户服务中的聊天机器人，识别投资机会或执行交易，增强信贷模型或作出贷款决策，以及识别和预防欺诈。某一行业或企业采用不同技术的程度，受到多种因素影响，其中包括企业自身开展内部研发的资金能力以及监管要求。

金融行业政策制定者面临相互竞争的压力。金融服务提供者和技术公司很可能继续采用和推广 AI/ML，以节省时间和成本，并提升可获得性、准确性和合规能力。然而，偏见、系统性风险和市场操纵可能性、可负担性以及就业影响等挑战和风险仍然存在。现

有监管结构是否足够，或者是否需要建立更贴合新技术能力的监管框架，已经成为重要问题。如果立法机关考虑金融领域 AI/ML 的制度框架，行业和消费者都会期待其在创新收益与既有及潜在风险之间取得平衡。

引言

人工智能是一个总称，用于描述把计算机和机器编程为像人一样思考和运行的过程。机器学习是人工智能的一个子集，指计算机和程序可以在最少人工干预下运行，并且在某些情况下能够学习或自我更新。过去几年发生的多个事件，提高了 AI/ML 及其在金融服务交付中作用的关注度。OpenAI 于 2022 年推出大型语言模型 ChatGPT，是 AI/ML 技术直接触达广大公众的少见时刻。虽然围绕 AI/ML 的发展势头至少已经持续了十多年，但多个重要技术的普及，使人们把注意力集中到各类应用之上，其中也包括金融领域的应用。

AI/ML 正在加速金融业长期持续的转变，即从客户所在社区中的面对面互动，转向由高级算法驱动的线上产品和服务。这些产品和服务几乎不需要人工互动，并且可以在任何地点发生。相关应用可能给金融机构及其客户带来收益。与人工操作或使用早期技术构建的模型相比，AI/ML 提供金融服务可能因为速度和效率而更快、更便宜；由于能够分析替代数据并发现潜在联系，它们还可能把服务扩展到更多个人。金融行业使用技术并不是新现象，其采用 AI/ML 的某些方面也与以往技术周期相似。

采用这类技术同样带来挑战和风险。AI 可能消除金融服务中的某些人为偏见，但也可能引入或加剧偏见。某些 AI 市场特定因素，例如使用相同数据训练模型，以及技术和技术能力的集中，可能鼓励单一化或类似羊群效应的行为，从而引入系统性风险。某些模型可以在缺乏人工监督的情况下自行学习，这也可能使其从事市场操纵、利益冲突或其他违法活动。

本报告广泛讨论 AI/ML 在金融服务中的作用。首先，它简要介绍若干技术术语；其次，讨论金融行业使用 AI/ML 的动机；随后，概述该技术在金融中的若干应用；最后，报告讨论监管框架，并考虑包括适用监管在内的多个政策问题。

技术概览

AI 是一个宽泛标签，指赋予计算机系统学习新概念或任务、进行推理并对世界得出有用结论能力的技术。早期 AI 在很大程度上依赖开发者，开发者需要用一系列确定性规则，例如 if-then 语句，并结合大量专业内容来编写机器。此类技术要求问题或命令采用特定语法，并只能对已经被编程的主题作出回应。当遇到没有被编程的情境时，它就会停滞。这种依赖性在当前分类中很可能不足以被认定为 AI，并且正在逐渐消失。有观点认为，现在大多数 AI 应用都是 ML，因此二者在实际使用中往往具有相近含义。

ML 是 AI 的一个子集，重点在于构建能够通过经验自动改进的计算机系统，并识别所有学习系统中固有的某些特征，这些学习系统可以包括计算机、人类和组织。ML 试图赋予计算机在无需重新编程的情况下学习和变化的能力。它使用算法，即为解决特定问题而设计的一系列步骤，并在不同程度、通常是很少甚至没有人工互动的情况下自动改进。换言之，与早期 AI 不同，ML 具有适应性。ML 系统旨在确定变量之间的关系，或识别大型数据集中的模式。不过，ML 并不是某一个离散的计算机应用，甚至也不是一种单一技术。它是一组系统，部分由训练模型所使用的信息类型以及人工训练者参与程度来定义。

计算和通信技术的发展极大促进了 AI/ML，金融行业对相关技术的使用也包括高性能计算，其中涵盖硬件、云计算、边缘计算和通信技术。不过，金融行业采用 AI/ML 还取决于数据可得性，包括传统金融决策中不使用的替代数据，以及处理和分析非结构化数

据的能力。这些技术进步与海量数据积累同步发生，而海量数据需要可扩展架构来实现高效存储、处理和分析。

今天某些技术的概念来源可以追溯到几十年前，而它们如今正在得到进一步发展。无监督学习处理无标签数据的能力，可以识别程序员原本没有想到去寻找的关系，例如市场互动之间的关系或贷款申请之间的关系。同样，对以往无法分析的非结构化数据和替代数据进行处理的能力，例如图像和社交媒体帖子，也使其能够进入分析过程。

大数据

大数据既指一种信息，也指这种信息的使用方式。它指具有容量、速度、多样性或可变性等特征的大型数据集，并需要可扩展架构来存储、管理和分析。该术语通常也指管理者、公司和行业使用数据来辅助商业决策。

ML 的一些进步直接归因于处理更多数据的能力。金融中使用的具体大数据取决于行业或使用场景。例如，一些公司在决定是否发放贷款时，可能使用包括社交媒体在内的大数据。在资本市场和交易环境中，大数据可能包括来自不同证券交易所的连续数据流，其中包含股票价格、更广泛的市场波动以及交易历史。这一环境中的数据还可能包括影响股票表现的公司层面数据，例如财务报表以及季度和年度监管文件。

替代数据指以往没有被用于金融决策的信息。在消费者信贷承销中，替代数据是指以往没有用于确定信用评分的非信贷数据，例如租金和水电费支付记录。银行、信用卡公司和支付公司还可能在欺诈检测和反洗钱活动中使用替代数据和非结构化数据，例如交易信息、位置数据或 GPS 数据。资产管理者和其他资本市场参与者也可能使用零售商停车场的 GPS 和卫星图像，或社交媒体提及信息，以洞察公司表现。不同形式的数据及其所使用的模型的可得性，扩展了金融企业使用二者的方式。

近期对 AI 的关注，伴随着各行业对 AI/ML 技术投资、开发和采用的增加，金融行业也包括在内。根据某银行研究，美国 AI 投资预计将在 2025 年升至约 1000 亿美元，全球投资接近 2000 亿美元。值得注意的是，传统金融机构很可能进行广泛投资，而面向金融的初创企业也吸引了风险投资机构的资金。

金融与人工智能

根据经济合作与发展组织的说法，AI 正在推动数字化转型，并已在金融领域快速部署。金融业采用这种新技术并不令人意外，因为金融行业常常与前沿技术相联系。然而，该行业也以使用陈旧技术，以及与新技术相冲突的传统实践相联系。过去，个人银行业务植根于社区关系，并通过面对面方式完成；如今，几乎所有交易都可以在线完成。在资本市场交易中，电子交易基本消除了必须在交易所现场通过人与人方式进行交易的需要。与此同时，AI/ML 的使用，类似于更广泛金融科技的采用，也会影响监管，因为监管强调行业透明性、可解释性和公平性。因此，金融领域采用 AI/ML 的过程，发生在技术及其能力发展可能快于社会和监管预期的背景下。

金融部门采用 AI 的动机并不新鲜。金融业采用 AI 曾被拿来与净现值计算的发展、自动柜员机的发明、用于个人理财规划的专家系统以及自动化交易进行比较，这些都曾经使用不同技术。

除了总体上倾向于使用技术之外，金融行业的若干特征也使其成为 AI/ML 的适宜试验场。信息处理是金融市场的核心功能。例如，AI/ML 通常需要大量数据来训练系统。数据是各金融部门的典型特征。该行业从经济指标、金融市场、个人消费者和企业、支付与交易等无数来源积累数据。此外，金融行业由不同子行业构成，具有多样功能、服务和产品，但金融问题通常是可量化的，其解决方案常被简化为在既定风险容忍度和其

他若干可量化变量约束下最大化收益。因此，金融行业既生成大量数据，也为 AI/ML 技术提供运行环境。

AI/ML 还可能提高金融交易发生的速度。例如，高频交易也越来越自动化和程序化，它允许交易在极短时间内发生，其速度是人类分析或执行都无法达到的。同样，如果金融机构使用 AI/ML 来判断是否发放贷款，这种速度也可能以多种方式体现出来。AI/ML 带来的速度可能使金融机构更快更新其贷款模型。贷款机构还可以把基于 AI/ML 的贷款模型与数字平台和网络界面集成，以更快向客户提供决策。

AI/ML 在金融中的应用

金融行业和市场参与者把 AI 用于多种功能。这些功能具有多重目的，包括提高效率、辅助决策、产生分析和预测、增加盈利能力、管理风险以及进行信贷承销。它们既可以用于支持性职能，也可以用于前台职能。后台应用可能包括交易后处理、交易盈亏核对和数据分析。其他应用可能包括监管合规职能，例如了解你的客户检查、客户身份识别项目、反洗钱和反恐融资项目以及反欺诈。AI 还可以辅助资产配置、智能投顾和交易执行。金融企业采取的许多行动都是多步骤且紧密相关的，AI 可以用于业务的不同部分或流程阶段，并从中受益。

信贷承销，即评估潜在借款人违约风险，是银行和其他金融机构的主要业务之一。发放贷款的金融机构已经通过电子数据等方式，使贷款决策方法自动化。对于消费者贷款，这些方法通常需要使用一个称为评分模型的数学公式来确定消费者信用评分，并进一步决定企业是否发放贷款以及采用何种利率。账单支付历史、未偿债务、未结清贷款和账户等多种因素都会影响信用评分。不过，近来一些金融公司开始考虑使用 AI/ML 来增强或替代传统信用评分。

贷款机构已经使用预测模型几十年。传统上，统计回归方法使用来自信用报告机构的数据，为不同变量分配权重，以帮助预测申请人是否会贷款违约，并确定按时还款的可能性。最近，企业可能使用基于 ML 的度量方法，因为这类方法能够分析大量且不同类型的数据，包括交易数据，并识别传统模型中不可见的其他重要关系。不同机构采用 AI/ML 进行信贷承销的情况可能不同。各种监管要求可能使银行在使用该技术作出信贷承销决策时更加谨慎。一般而言，由于非银行金融科技机构面临的监管要求和监管方式可能不同于银行，它们可能更愿意使用 AI/ML 技术。

聊天机器人是与人在线互动的计算机程序，可以通过文本和语音命令模拟人与人的对话。银行聊天机器人能够全天候提供即时帮助，减少等待时间，回应客户咨询，提供账户余额和交易历史信息，并引导用户完成各种银行流程。根据一项估计，银行中使用的聊天机器人每年能够带来可观的成本节约。

监管科技是指受监管金融机构使用技术来满足各种监管、合规和数据报告职能。监管科技已经存在约十年。它最初侧重于与新客户 onboarding 和客户识别有关的监管合规，近来则被认为在反洗钱、反恐融资、欺诈预防、风险管理、压力测试以及微观和宏观审慎报告方面发挥重要作用。例如，银行和其他金融机构使用机器人流程自动化，帮助其遵守报告要求，并定期重新填充所需数据。在许多情况下，银行和其他金融机构采用监管科技，可能需要与拥有该领域专业能力的第三方提供商合作。

由 AI/ML 支持的一种重要监管科技形式，是为银行和其他金融机构检测、预防并报告未经授权和非法金融活动。转向 AI/ML 解决方案的原因，在于其方法更具适应性，并且能够利用更多类型的数据。银行和其他金融机构必须检测、预防并报告未经授权和非法金融活动，使用 AI/ML 处理这一问题的金融机构数量也在增加。银行和其他金融机构

可以用其生成的大量消费者行为数据训练模型，使 ML 模型学习欺诈模式，并在实践中检测欺诈行为。同样，在反洗钱价值链中使用该技术也可能带来收益，包括在客户准入、客户筛查以及交易监测中获得特别且直接的好处。此外，研究者认为，该技术还有助于减少误报，使银行能够把资源投入到真实欺诈事件中。

自然语言处理是 AI 的一个分支，它训练计算机像人类一样理解和处理语言。NLP 可用于开发大型语言模型，使其根据提示生成文章或计算机代码，其方法是基于某些概率预测词语序列中的下一个词。这项技术支持金融中的多种应用，包括驱动聊天机器人、检索企业自身的大量数据，以及编写代码。大型语言模型是人工神经网络的例子，它们试图通过把神经元复制为数学表征，并把这些人工神经元组织为层或簇，来反映人脑的运行方式。除了金融专用应用之外，大型语言模型，尤其是 ChatGPT，也让普通公众首次获得了具体且可直接使用的 AI 示例，而这些能力此前通常只为专业人员所用。